

JoanMarie Leone

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EDUCATION

Binghamton University, State University of New York, Thomas J. Watson College of Engineering and Applied Science

Master of Science in Computer Science, Artificial Intelligence Track

August 2025

Cumulative GPA: 3.88/4.00

Relevant Coursework: Artificial Intelligence, Data Mining, Intelligent Mobile Robotics, Science of Cyber Security

Bachelor of Science in Computer Science

May 2024

Cumulative GPA: 3.58/4.00

Relevant Coursework: Machine Learning, Data Structures and Algorithms, Programming Languages, Operating Systems

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, Java, X-86 Assembly

Software: Git, GitHub, Visual Studio Code, Unix, Jupyter Notebook, Logisim, Linux, ROS2, JFlap, Android Studio, IsaacSim

Additional: Azure, PostgreSQL

PROJECT EXPERIENCE

Black Box Prompt Optimization Defense for Prompt Injection Attacks, Independent Project *June 2025 - August 2025*

- Introduced Black Box Prompt Optimization (BPO) as a defense against Prompt Injection Attacks
- Created custom datasets pairing prompt injection variants with safe (good) and unsafe (bad) responses
- Trained a prompt optimizer to generate refined prompts that prevent malicious instructions from being executed
- Laid groundwork for future research utilizing BPO as a defense towards PI attacks

Crawler and Analysis on Online Music Discourse, Independent Project

September 2024 - December 2024

- Built two custom continuous crawlers to obtain posts from Reddit music subreddits and the 4Chan music board
- Used online datasets to parse and identify music artists in large sets of online posts
- Handled and organized a large amount of data using PostgreSQL and Python libraries such as Pandas
- Analyzed and evaluated trends in artists' mentions, frequency, and toxicity scores of posts mentioning artists

Return-To-Libc Attack Demonstration, Independent Project

September 2023 - December 2023

- Created a susceptible executable with C and reverse-engineered the file binary using Cutter
- Constructed a Python exploit file, including ROP gadgets and libc function addresses in the binary to expose libc
- Implemented and analyzed the effectiveness of different detection, prevention, and deception methods

Xv6 Multi-Level Feedback Queue Process Scheduler Implementation, Team Project

September 2023 - November 2023

- Utilized knowledge of operating systems to build a multi-level process scheduler from baseline xv6 source code
- Efficiently communicated and collaborated with two other students to complete the MLFQ scheduler
- Applied different scheduling policies, including round-robin and priority-based for process execution
- Created new system call functions in baseline xv6 to aid in implementing the multi-level queue and scheduling policies

Hardware Trojan Detection Technical Research Paper, Independent Project

March 2022 - May 2022

- Researched different forms of Hardware Trojan detection algorithms, focusing on run-time monitoring methods
- Compared the reliability, coverage, resource consumption, and power consumption of HT detection algorithms

LEADERSHIP AND TEAMWORK EXPERIENCE

The Association For Computing Machinery, Member

January 2022 – January 2024

- Partook in coding competitions in teams of one or two, tackling easy to challenging coding problems in timed environment
- Learned and sharpened knowledge of various computer science topics by attending weekly ACM presentations

HackBU (Binghamton University), Member

January 2022 – January 2024

- Explored different real-world applications of computer science in weekly meetings
- Strengthened project management, programming, and debugging skills in workshops

WORK EXPERIENCE

Sag Harbor Pierson Middle/High School, Senior Office Assistant | Sag Harbor, NY

January 2026 - Present

- Balance and prioritize multiple tasks simultaneously, ensuring all tasks are completed accurately and on schedule
- Maintain and organize a digital and physical filing system including student records, transcripts, and confidential records
- Serve as the primary contact for students, parents, and staff in the guidance office, providing professional assistance

Lombardi's Love Lane Market, Deli Employee | Mattituck, NY

June 2021 - December 2025

- Safely operated meat slicers and other sharp or hot equipment while preparing high-quality deli products in lively market.
- Managed, coordinated, and completed catering orders before strict deadlines

Long Island Aquarium and Exhibition Center, Guest Services Employee | Riverhead, NY

June 2017 - August 2019

- Operated ride machinery while ensuring safe performance and strict adherence to Aquarium safety guidelines
- Honed personal relations and marketing skills by selling products and event tickets to customers